

MARKET SHOCK

SHOCK TYPE

Equity Flash Crash

REFERENCE EVENT

May 6, 2010 — Synthetic Derivative

SEVERITY (DRAWDOWN)

-18.7%

DURATION

4h 23m

RECOVERY (EST.)

-90 min

SHOCK DESCRIPTION

Synthetic equity flash crash derived from the May 6, 2010 event. Intraday drawdown of -18.7% on an S&P 500 equivalent index over a 4h 23m window, followed by partial recovery. Market microstructure conditions replicated at 1ms tick resolution.



DECISION SYSTEMS UNDER ASSESSMENT

Select the AI-enabled financial decision systems to assess. All selected systems will receive the identical shock stimulus under controlled, isolated conditions.

☒

Citadel AMS v3.2
Automated Market-Making

✓

☒

BlackRock Aladdin Risk Engine
Portfolio Risk Management

✓

☒

Two Sigma Venn Platform
Quantitative Trading System

✓

☒

JP Morgan LOXM
Execution Algorithm

✓

☒

Virtu Financial Execution Engine
HFT Market Making

✓

☐

Man AHL Signature Fund
Systematic Macro

5 of 6 systems selected

ASSESSMENT PARAMETERS

ASSESSMENT TYPE

Convergence Analysis

CONTROLLED COMPARISON

Isolated shock injection

OBSERVATION WINDOW

T-30m to T+120m

SAMPLING INTERVALS

T+0, T+5, T+15, T+30, T+60m

MARKET MODEL

Synthetic microstructure v4.1

☐

This assessment runs in a sandboxed environment. No live market data or production systems are accessed. Results are f

Created with Figma Make

Remix

...

×

EVIDENCE SOURCES

☐

S&P 500 tick data — May 6, 2010 (CRSP)

×

☐

VIX options chain — 09:30-16:00 ET

×

☐

Order book reconstruction — NYSE/NASDAQ

×

☐

SEC/CFTC Joint Report — September 2010

×

☐

ESMA AI Convergence Study — 2023

×

+ Add evidence source

ASSESSMENT BOUNDARIES

IN SCOPE

Domestic equity markets (US). S&P 500 constituents. Primary exchange order flow only.

OUT OF SCOPE

FX spillover. Credit market contagion. Non-US equity markets. Off-exchange/dark pool venues.

REGULATORY FRAMEWORK

MiFID II Art. 17, 48; SEC Rule 15c3-5; ESMA AI Supervisory Guidance 2024